Descriptive Statistics

Principles of Information Quality

INFQ 7303

John R. Talburt

Descriptive vs. Inferential Statistics

- Data

  Set of values that are collected, recorded, and organized, and from which conclusions are drawn.

- Descriptive statistics

  About summarizing the collected data (measurements)

- Inferential statistics

  Trying to reach conclusions that extend beyond the immediate data alone

- What are the data telling me?

Population

- **Population** - The set of all entities that could be considered within the context of the problem under analysis.
- Examples:
  - All U.S. citizens
  - All customer sales records for sales in December 2017
  - All female students enrolled in Big State University.

Population Sample

- **Sample**
  A selected portion of the population.
- **Purpose of Sampling**
  Provides a way to understand the characteristics of a population when examining every member of the population is impractical or impossible
- **Goal of Sampling**
  Minimize the difference between the characteristics of the sample and the population from which the sample was taken

Parameter versus Sample Statistic

- **Parameter of the population** - Any value or measure that describes some characteristic of the data over the entire population.
- Traditionally represented by Greek letters
  - *Average height of all female students enrolled at Big State University.*

- **Sample statistic** - Any value or measure that describes some characteristic of the data in the sample.
- Traditionally represented by Latin letters
  - *Average height of all the female students taking the class Intro to IQ at Big State University.*

Sampling Error

- The difference between a population parameter and its corresponding sample statistic

- If we knew the true value of the population parameter, then we would not need a sample to estimate its value

- The way we assure a sample statistic is close to the value of the population parameter is to create the sample by "random selection", i.e. a random sample

Four Types of Random Sampling

1. Simple Random Sampling
2. Systematic Sampling
3. Stratified Sampling
4. Cluster Sampling

Simple Random Sampling

- Items are chosen at random from the entire population
- Assign a unique number to every member of the population, then generate a list of random numbers to select members
- Assigning unique numbers to the population members is not always practical
- Example: Lightbulbs coming off the end of an assembly line

Systematic Sampling

- Sample members are selected from the population at preset intervals
- Inspect every 50th lightbulb
- Inspection every 100th row in a database table
- Sometimes called “nth-ing”

Stratified Sampling

- Divide the population is divided into mutually exclusive groups based on one or more relevant population features.
- The groups are called “strata”, each group is a “stratum”
- Draw a random sample from each stratum.
- The sample statistic of each stratum sample is weighted by the size of the stratum relative to the entire population

Stratified Sampling Example

- Estimating the error rate in student degree plans
- Historically, errors are more likely in degree plans for students with transfer credits and students with change in major

Stratum	Transfer	Major Change	Count	Error in Sample	Stratum Weight	Adjusted Error
1	NO	NO	4,000	1%	0.80	0.80
2	YES	NO	600	3%	0.12	0.36
3	NO	YES	300	2%	0.06	0.12
4	YES	YES	100	7%	0.02	0.14
		Total	5,000			1.42%

Cluster Sampling

- Similar to Stratified Sampling, but more subjective
- Used in the case of a dispersed population, when excessively expensive to reach each member.
- The total population is divided into clusters, where each of which should be a small-scale version of the total population (subjective)
- Random sampling is then used on the set of clusters to choose which clusters to include in the study.
- All selected clusters are given equal weight
  i.e. Estimated population parameter is average of sample statistics

Descriptive Statistics

Organizing and summarizing collected data

Trade-Off

Grouped and summarized data

- Loses much of the original detail,

but in exchange, a

- Clearer overall picture can be obtained of some of the essential characteristics of the raw data.

Example Data

- Random sample of salaries in a company to the nearest thousands of dollars
- Sample of 40 Values

62, 54, 75, 66, 65, 55, 65, 89, 67, 52,
57, 70, 63, 78, 85, 92, 52, 55, 67, 69,
88, 82, 58, 90, 95, 79, 85, 82, 75, 70,
67, 56, 70, 67, 75, 69, 67, 75, 79, 90

Value Frequency

- 23 distinct values
- Not that helpful in this case

52	2
54	1
55	2
56	1
57	1
58	1
62	1
63	1
65	2
66	1
67	5
69	2
70	3
75	4
78	1
79	2
82	2
85	2
88	1
89	1
90	2
92	1
95	1

Frequency Distribution

- Data is organized in terms of categories or classes.
- Class frequency - The number of data items in each class.
- Frequency distribution - A data table together with the corresponding frequencies for each class.
  - May also show cumulative frequency
  - And relative frequency (percentage of total) and cumulative percentage

Frequency Table for Salary Survey

Class Interval	Frequency	Cumulative Frequency	Relative Frequency	Cumulative Relative Frequency
50–59	8	8	0.20	0.20
60–69	12	20	0.30	0.50
70–79	10	30	0.25	0.75
80–89	6	36	0.15	0.90
90–99	4	40	0.10	1.00

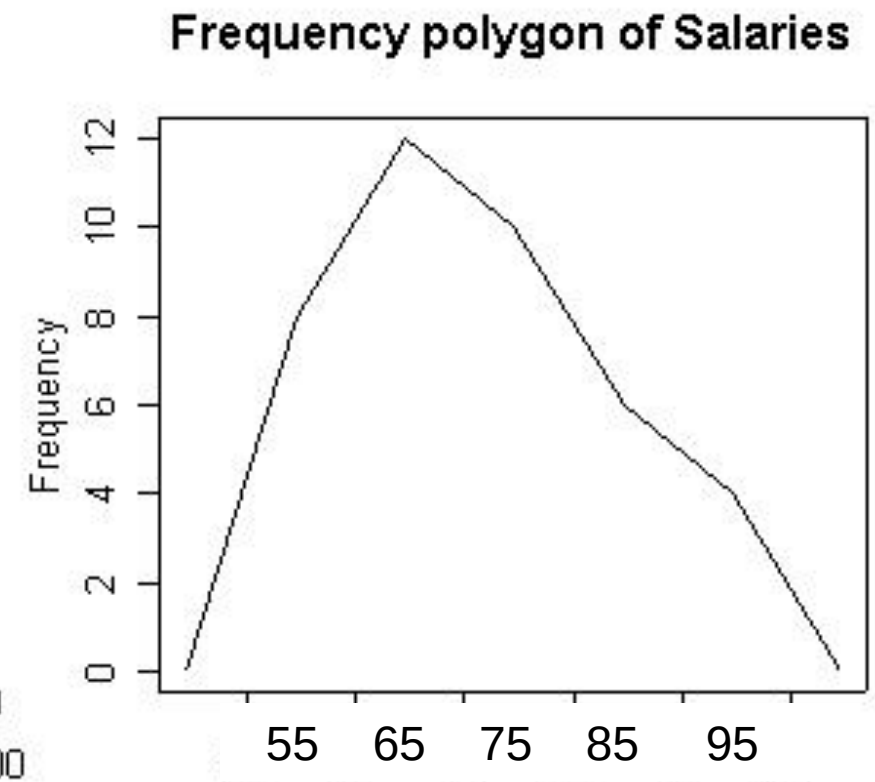
Histogram and Frequency Polygon

- **Histogram** - A bar chart of a frequency distribution, with bar widths equal to the class interval sizes and numbers of observations listed along the vertical axis.

- **Frequency Polygon** - A line graph of the frequency distribution where the midpoint of each class interval is displayed along the horizontal axis.

Example of a Histogram and Frequency Polygon





Measures of Central Tendency

- Quantifies the location of the middle or the center of a distribution.

- Three measures of central tendency include:
  - **Mean** (Average Value)
  - **Median** (Middle Value)
  - **Mode** (Most Common Value)

Mean

- Arithmetic mean (average) - usually defined as ,

$$Mean = \frac{\sum_{i=1}^{N}(X_i)}{N}$$

- where $\mathbf{X_i}$ are the **N** data values assumed by variable **X**.

Mean

- Mean of population of size N is $\mu = \frac{\sum_{i=1}^{N}(X_i)}{N}$
- Sample mean of size n is $\bar{X} = \frac{\sum_{i=1}^{n}(X_i)}{n}$
- Salary Sample Mean = 71.425
- Excel Function “AVERAGE( )”

Median

- **Median** of a group of values arranged in order of magnitude
  - If odd number of values, median is the value of the middle item
  - If even number of values, median is the average of the two middle items
  - Salary Sample Median = 69.5
  - Excel Function "MEDIAN( )"

Mode

- **Mode** - the value that occurs most frequently in the data set.
  - Salary Sample Mode = 67
  - Excel Function "MODE( )"
  - Not always helpful
    - May not have a mode
    - May have more than one mode

Question

- Given the following sample of five values {3, 4, 4, 8, 11}, what are the:
  - Mean?
  - Median?
  - Mode?

Answers

- Given the following sample of five values $\{3, 4, 4, 8, 11\}$,
  - Mean = 6
  - Median = 4
  - Mode = 4

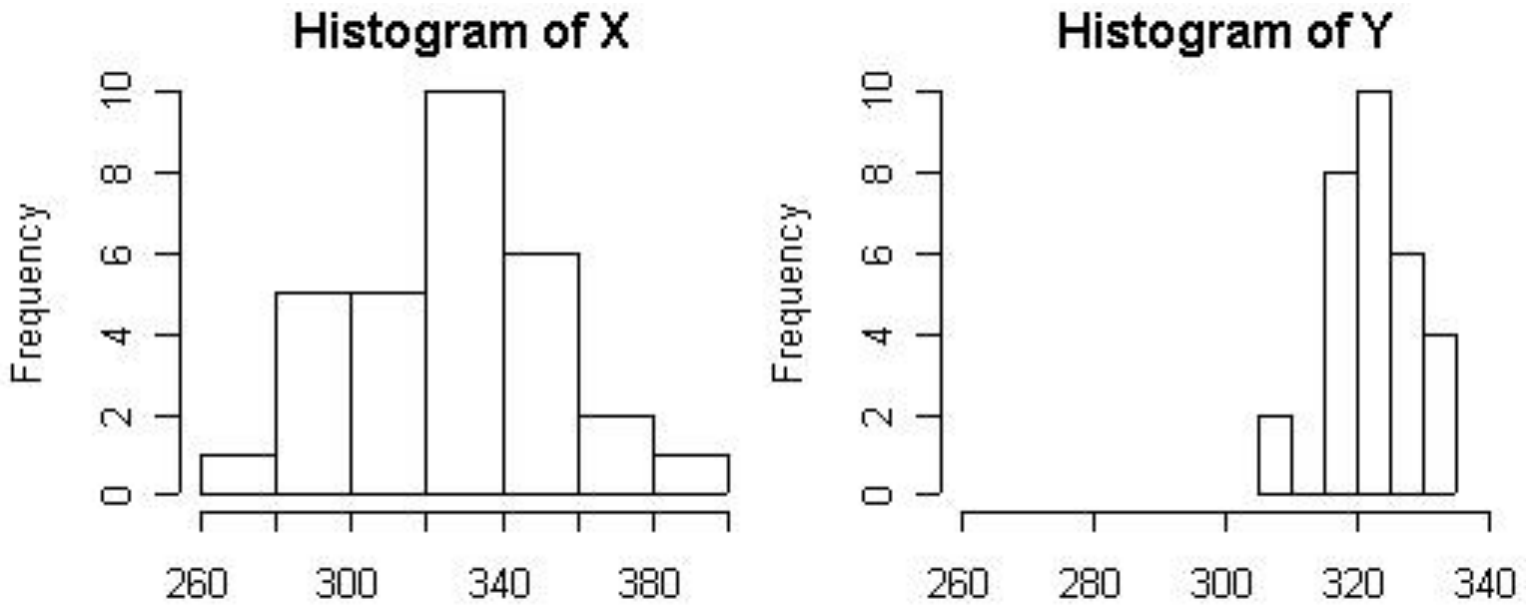
Sample Data Sets

X = 349, 324, 289, 339, 330, 306,
306, 354, 302, 303, 334, 343,
337, 349, 335, 282, 307, 351,
328, 298, 375, 294, 296, 325,
353, 368, 265, 336, 337, 395

Y = 322, 320, 330, 323, 320, 322,
330, 327, 327, 319, 331, 327,
322, 326, 330, 332, 326, 325,
317, 320, 329, 321, 322, 323,
323, 327, 337, 332, 325, 322

- Seem like similar values, but difficult to tell

Spread of Distributions



- Very different dispersion of values
- X from 289 to 395
- Y from 317 to 337

Measures of Spread

- Based on calculations of distances between points in a distribution.
- The degree of variability in a numerical data set.
- Common measures of spread include
  - Range
  - Standard deviation
  - Variance

Range

- The distance between the highest and the lowest measurements in a distribution.

- It is calculated by subtracting the smallest value (minimum value) from the largest value (maximum value)

Question

- What is the Range of the Salary Sample data?

62, 54, 75, 66, 65, 55, 65, 89, 67, 52,
57, 70, 63, 78, 85, 92, 52, 55, 67, 69,
88, 82, 58, 90, 95, 79, 85, 82, 75, 70,
67, 56, 70, 67, 75, 69, 67, 75, 79, 90

Answer

- Find the largest and smallest values

62, 54, 75, 66, 65, 55, 65, 89, 67, 52,
57, 70, 63, 78, 85, 92, 52, 55, 67, 69,
88, 82, 58, 90, 95, 79, 85, 82, 75, 70,
67, 56, 70, 67, 75, 69, 67, 75, 79, 90

- Range = 95-52 = 43

Standard Deviation

- Whereas the Range only consider two values in the data set, the Standard Deviation measures the distance of every data elements with respect to their mean value.

- If the data set is tightly grouped the standard deviation will be small

- If the data is widely dispersed the standard deviation will be large

Standard Deviation of a Population

- The standard deviation of the entire population is denoted by **σ** (lower case Greek letter sigma) and has the following formula:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(X_i - \mu)^2}{N}}$$

Standard Deviation- Sample

- The standard deviation of the sample of size ***n***, denoted by ***s***, has the following formula:

$$s = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{(n-1)}}$$

Question

- What is standard deviation of $\{2, 4, 4, 8, 12\}$ as a complete population?

- What is standard deviation of $\{2, 4, 4, 8, 12\}$ as a sample of a larger population?

Answers

$2 + 4 + 4 + 8 + 12 = 30$

$\text{Avg} = 30/5 = 6$

$(2\text{-}6)(2\text{-}6)+(4\text{-}6)(4\text{-}6)+(4\text{-}6)(4\text{-}6)+(8\text{-}6)(8\text{-}6)+(12\text{-}6)(12\text{-}6)$

$=16 + 4 + 4 + 4 + 36 = 64$

$\text{STD as Population} = \sqrt{\frac{64}{5}} = \frac{8}{\sqrt{5}} = \frac{8\sqrt{5}}{5} = 3.578$

$\text{STD as Sample} = \sqrt{\frac{64}{4}} = \sqrt{16} = 4.00$

Variance of a Set of Data

- The square of the standard deviation
- Denoted σ^2 for a population
- Denoted by s^2 for a sample

Z-Score

- Distance from the mean in units of standard deviation

$$Z - Score\ of\ X = \frac{(X - \mu)}{\sigma}$$

- A way to "standardize" measurements
  - The Z-Score of μ = 0
  - The Z-Score of a value greater than μ is positive
  - The Z-Score of a value less than μ is negative

Values Normalized by Z-Score

	SQ FT	Norm
1	500	-0.97
2	1,000	-0.65
3	2,000	0
4	4,500	1.62
Avg	2,000	
StDv	1,541	

	Age	Norm
1	1	-0.85
2	10	-0.62
3	25	-0.23
4	100	1.69
Avg	34	
StDv	39	

Questions and Discussion